# EE3110 - Probability Foundations for Electrical Engineers
# Tutorial - 7

1. Let $f$ be a probability function for a distribution. Suppose that $f(x) = 0$ for $x \notin [0,1]$. Prove that the variance of this distribution is at most $1/4$. Next, prove that this bound is tight by providing a distribution which achieves the upper bound.

**Solution:** Let $X$ be a random variable supported on $[0,1]$, with mean $\mu = \mathbb{E}[X] \in [0,1]$. Then the variance is

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mu^2.$$

Since $0 \le X \le 1$, we have $X^2 \le X$ for all values of $X$. Taking expectations gives

$$\mathbb{E}[X^2] \le \mathbb{E}[X] = \mu.$$

Therefore,

$$\mathrm{Var}(X) \le \mu - \mu^2 = \mu(1-\mu).$$

Note that this is maximized at $\mu = \frac{1}{2}$ for $\mu \in [0,1]$ and the maximum value is $\frac{1}{4}$.

Now consider a Bernoulli random variable $Y$ supported on $\{0,1\}$ with $\mathbb{P}(Y = 1) = \frac{1}{2}$. For this distribution,

$$\mathrm{Var}(Y) = \frac{1}{4},$$

which achieves the desired bound.

2. Let $X_n$ be a random variable with the binomial distribution $\mathrm{Bin}(n, p_n)$. Assume that $\lim_{n\to\infty} np_n = \lambda \in [0,\infty)$. Prove that the moment generating function (m.g.f.) of $X_n$ converges pointwise (i.e. for every real $t$) to the m.g.f. of the Poisson distribution with mean $\lambda$.

**Solution:** For $X_n \sim \mathrm{Bin}(n, p_n)$, its m.g.f. is

$$M_{X_n}(t) = \mathbb{E}\big[e^{tX_n}\big] = \big(1 - p_n + p_n e^t\big)^n = \big(1 + p_n(e^t - 1)\big)^n.$$

Let $a_n := np_n$ so that $a_n \to \lambda$. Then

$$M_{X_n}(t) = \left(1 + \frac{a_n(e^t - 1)}{n}\right)^n.$$

Taking logarithms,

$$\log M_{X_n}(t) = n \log\left(1 + \frac{a_n(e^t - 1)}{n}\right).$$

Using the expansion $\log(1+u) = u + o(u)$ as $u \to 0$, we get

$$\log M_{X_n}(t) = a_n(e^t - 1) + o(1) \xrightarrow[n\to\infty]{} \lambda(e^t - 1).$$

Exponentiating and using the continuity of the exponential function yields

$$M_{X_n}(t) \xrightarrow[n\to\infty]{} \exp\big(\lambda(e^t - 1)\big),$$

which is precisely the m.g.f. of the Poisson($\lambda$) distribution. Hence the m.g.f. of $X_n$ converges to that of Poisson($\lambda$).

3. Let $X_1, X_2, ...$ be i.i.d positive random variables with mean $a > 0$. Let $Y_1, Y_2, ...$ be i.i.d positive random variables with mean $b > 0$. Show that

$$\frac{X_1 + X_2 + ... + X_n}{Y_1 + Y_2 + ... + Y_n} \to c$$

where $c > 0$, with probability 1. What is the value of $c$? Does it matter whether the $X_i's$ are independent of the $Y_j's$?

**Solution:** By the Law of Large Numbers (LoLN),

$$\frac{X_1 + X_2 + ... + X_n}{n} \to a$$

with probability 1, and

$$\frac{Y_1 + Y_2 + ... + Y_n}{n} \to b$$

with probability 1 as $n \to \infty$. Since the above events occur with probability 1 individually, their intersection must also occur with probability 1. So with probability 1, both the convergence of $X_i's$ and convergence of $Y_i's$ occur, and

$$\frac{X_1 + X_2 + ... + X_n}{Y_1 + Y_2 + ... + Y_n} \to \frac{a}{b}$$

with probability 1. So, $c = a/b$. Also note, it was not necessary to assume that the $X_i's$ are independent of $Y_j's$.

4. Let $U_1, U_2, ..., U_{60}$ be i.i.d Unif(0, 1), and $X = U_1 + U_2 + ... + U_{60}$. Approximate the distribution of $X$ with appropriate parameters, and give a simple but accurate approximation for $P(X > 17)$. Justify your answer.

**Solution:** We have $\mathbb{E}[X] = 30$, and Var(X) $= 60/12 = 5$. By Central Limit Theorem (CLT), the distribution of $X$ is approximately $\mathcal{N}(30, 5)$. Now,

$$P(X > 17) = 1 - P(X \leq 17) = 1 - P\left(\frac{X - 30}{\sqrt{5}} \leq \frac{-13}{\sqrt{5}}\right) \approx 1 - \Phi\left(\frac{-13}{\sqrt{5}}\right) = \Phi\left(\frac{13}{\sqrt{5}}\right)$$

which is a value very close to 1.

5. (a) Consider i.i.d. Poisson($\lambda$) r.v.s $X_1, X_2, \ldots$. The MGF of $X_j$ is $M(t) = e^{\lambda(e^t - 1)}$. Find the MGF $M_n(t)$ of the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} X_j.$$

(b) Find the limit of $M_n(t)$ as $n \to \infty$.

**Solution:**

(a) The MGF is

$$E\left(e^{\frac{t}{n}(X_1+\cdots+X_n)}\right) = \left(E\left(e^{\frac{t}{n}X_1}\right)\right)^n = e^{n\lambda(e^{t/n}-1)},$$

since the $X_j$ are i.i.d. and $E(e^{\frac{t}{n}X_1})$ is the MGF of $X_1$ evaluated at $t/n$.

(b) By the law of large numbers, $\bar{X}_n \to \lambda$ with probability 1. The MGF of the constant $\lambda$ (viewed as an r.v. that always equals $\lambda$) is $e^{t\lambda}$. Thus,

$$M_n(t) \to e^{t\lambda} \quad \text{as } n \to \infty.$$

6. (a) Suppose you toss a fair coin 100 times. Find the probability that you end up with at least 90 heads using Markov inequality.
(b) Can you come up with a non-negative random variable $X$ with $E[X] = 50$ that achieves the upper bound above for $P(X > 90)$?
(c) For a general non negative random variable X that has a finite expectation, show that the equality case can always be achieved in the Markov Inequality. By doing so, you are showing that Markov Inequality cannot be improved in general.

**Solution:**

(a) Let X be the number of heads. Then X is Bin(100,0.5). Apply Markov inequality to get

$$\mathbb{P}(X \geq 90) \leq \frac{50}{50} = \frac{5}{9}$$

(b) Let X' be a random variable that takes value 90 with probability $\frac{5}{9}$ and 0 otherwise. This makes

$$\mathbb{P}(X \geq 90) = \mathbb{P}(X = 90) = \frac{5}{9}$$

(c) From part(b) we realise that for any constant c in the Markov inequality

$$\mathbb{P}(X \geq c) \leq \frac{\mathbb{E}(X)}{c}$$

we can define a random variable X' that takes value c with some probability p and value 0 otherwise. This achieves equality.

7. Let $X_1, X_2, \ldots, X_n$ be a sequence of i.i.d. random variables with mean $\mu = 10$ and variance $\sigma^2 = 25$. Define $S_n = \sum_{i=1}^n X_i$ and $\overline{X}_n = \frac{S_n}{n}$.
   1. According to the Central Limit Theorem, what is the distribution of $\overline{X}_n$ as $n \to \infty$?
   2. For $n = 100$, calculate the approximate probability $P(9.5 \leq \overline{X}_n \leq 10.5)$.

**Solution:**

(a) By the Central Limit Theorem (CLT), as $n \to \infty$, the sample mean $\overline{X}_n$ of i.i.d. random variables with mean $\mu$ and variance $\sigma^2$ is approximately normally distributed:

$$\overline{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

For this question, $\mu = 10$ and $\sigma^2 = 25$, so as $n \to \infty$,

$$\overline{X}_n \approx \mathcal{N}\left(10, \frac{25}{n}\right).$$

(b) With $n = 100$, the variance of $\overline{X}_n$ is $\frac{\sigma^2}{n} = \frac{25}{100} = 0.25$, so

$$\overline{X}_n \approx \mathcal{N}(10, 0.25).$$

To find $P(9.5 \leq \overline{X}_n \leq 10.5)$, we standardize this probability:

$$P(9.5 \leq \overline{X}_n \leq 10.5) = P\left(\frac{9.5 - 10}{\sqrt{0.25}} \leq Z \leq \frac{10.5 - 10}{\sqrt{0.25}}\right).$$

Calculating the bounds:

$$P(9.5 \leq \overline{X}_n \leq 10.5) = P(-1 \leq Z \leq 1).$$

Using the standard normal table, we find:

$$P(-1 \leq Z \leq 1) \approx 0.6826.$$

Thus, $P(9.5 \leq \overline{X}_n \leq 10.5) \approx 0.6826$.

8. The total number of students attending class on any given week is a random variable with mean 50. If the variance in a week's attendance is known to be 25, then the probability that the attendance in any week will be between 40 and 60 is at least _____ ?

**Solution:** Let $X$ be the number of students attending class in a week. Applying the Chebyshev inequality, we have

$$\mathbb{P}[|X - 50| \geq 10] \leq \frac{\sigma^2}{10^2} = \frac{1}{4}$$

Therefore,

$$\mathbb{P}[|X - 50| < 10] \geq 1 - \frac{1}{4} = \frac{3}{4}$$

9. A physicist makes 25 independent measurements of the specific gravity of a certain body. He knows that the limitations of his equipment are such that the standard deviation of each measurement is $\sigma$ units.

   (a) By using the Chebyshev inequality, find a lower bound for the probability that the average of his measurements will differ from the actual specific gravity of the body by less than $\sigma/4$ units.

   (b) By using the central limit theorem, find an approximate value for the probability in part (a).

**Solution:** Let $\bar{X}_n$ denote the average of the $n = 25$ independent measurements. We are given that each measurement has mean $\mu$ and standard deviation $\sigma$.

(a) By the **Chebyshev inequality**,

$$P(|\bar{X}_n - \mu| \geq k) \leq \frac{\text{Var}(\bar{X}_n)}{k^2}.$$

Now, $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} = \frac{\sigma^2}{25}$, and we take $k = \frac{\sigma}{4}$. Hence,

$$P(|\bar{X}_n - \mu| \geq \tfrac{\sigma}{4}) \leq \frac{\sigma^2/25}{(\sigma/4)^2} = \frac{1/25}{1/16} = \frac{16}{25}.$$

Therefore,

$$P(|\bar{X}_n - \mu| \leq \tfrac{\sigma}{4}) \geq 1 - \frac{16}{25} = 0.36.$$

(b) By the **Central Limit Theorem**, the standardized variable

$$Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = 5\,\frac{\bar{X}_n - \mu}{\sigma}$$

is approximately standard normal, i.e. $Z \sim N(0, 1)$.

Thus,

$$P(|\bar{X}_n - \mu| \leq \tfrac{\sigma}{4}) = P(|Z| \leq \tfrac{5}{4}) = 2\Phi(1.25) - 1.$$

From standard normal tables, $\Phi(1.25) \approx 0.8944$, giving

$$P(|\bar{X}_n - \mu| \leq \tfrac{\sigma}{4}) \approx 2(0.8944) - 1 = 0.7888.$$

| | |
|---|---|
| (a) | $P(\lvert\bar{X}_n - \mu\rvert \leq \frac{\sigma}{4}) \geq 0.36,$ |
| (b) | $P(\lvert\bar{X}_n - \mu\rvert \leq \frac{\sigma}{4}) \approx 0.7887.$ |

10. A political candidate is concerned about voter support in a large district. Her campaign manager claims that she has 52% support ($p = 0.52$). The candidate will only proceed with a new, expensive advertising campaign if she is convinced her support is less than 50%. They decide to conduct a poll of $n$ voters. They will launch the new campaign if the sample proportion of supporters, $\hat{p}$, is less than 50%. Assuming the campaign manager is correct (that the true support is $p = 0.52$), what is the **minimum number of voters** $n$ they must poll to ensure that the probability of *incorrectly* launching the new campaign (i.e., finding $\hat{p} < 0.50$) is **no more than** 1%?

Use the CLT to approximate probabilities as necessary. The critical value $Z$ corresponding to a 1% tail probability, $P(Z \leq z) = 0.01$, is $\mathbf{z_{crit} \approx -2.326}$.

**Solution:** Let $X$ = the **number** of supporters found in the sample of $n$ voters. The variable $X$ follows a Binomial distribution:

$$X \sim \text{Bin}(n, p = 0.52)$$

We need to find the minimum $n$ such that the probability of incorrectly launching the campaign is $\leq 1\%$. The condition for launching is $\hat{p} < 0.50$. Since $\hat{p} = \frac{X}{n}$, this is the same as $\frac{X}{n} < 0.50$, or $\mathbf{X < 0.50n}$.

We must find the smallest integer $n$ that satisfies:

$$P(X < 0.50n) \leq 0.01$$

**Central Limit Theorem (CLT)**

We approximate the discrete Binomial variable $X$ with a continuous Normal variable $Y \sim N(\mu, \sigma^2)$ where:

- **Mean ($\mu$):** $\mu = np = n(0.52) = \mathbf{0.52n}$
- **Variance ($\sigma^2$):** $\sigma^2 = np(1-p) = n(0.52)(1-0.52) = n(0.52)(0.48) = \mathbf{0.2496n}$
- **Standard Deviation ($\sigma$):** $\sigma = \sqrt{0.2496n} \approx \mathbf{0.4996\sqrt{n}}$

**Apply the Correction for Continuity**

We are approximating the **discrete** probability $P(X < 0.50n)$ with a **continuous** normal variable $Y$. The rule for "strictly less than" ($<$) is to subtract 0.5 from the boundary.

$$P(X < 0.50n) \approx P(Y \leq 0.50n - 0.5)$$

Our probability statement is now:

$$P(Y \leq 0.50n - 0.5) \leq 0.01$$

**Standardize to a Z-score**

We convert $Y$ into a standard normal variable $Z$ using the $Z$-score formula: $Z = \frac{\text{value} - \mu}{\sigma}$.

$$P\left(Z \leq \frac{(0.50n - 0.5) - (0.52n)}{\sqrt{0.2496n}}\right) \leq 0.01$$

Simplifying the numerator:

$$P\left(Z \leq \frac{-0.02n - 0.5}{\sqrt{0.2496n}}\right) \leq 0.01$$

**Critical Z-value**

From the hint provided in the question, we know the $Z$-score that corresponds to the bottom 1% of the curve is $z_{crit} \approx -2.326$.

Therefore, our inequality becomes:

$$\frac{-0.02n - 0.5}{\sqrt{0.2496n}} \leq -2.326$$

This is the main algebraic challenge.

$$\frac{-0.02n - 0.5}{\sqrt{0.2496}\sqrt{n}} \leq -2.326$$

$$\frac{-0.02n - 0.5}{0.4996\sqrt{n}} \leq -2.326$$

Multiply both sides by $0.4996\sqrt{n}$:

$$-0.02n - 0.5 \leq -2.326 \cdot (0.4996\sqrt{n})$$

$$-0.02n - 0.5 \leq -1.1619\sqrt{n}$$

Rearrange all terms to one side to form a quadratic-style inequality:

$$\mathbf{0.02n - 1.1619\sqrt{n} + 0.5 \geq 0}$$

**Solving the Inequality:** To solve this, we use a substitution. Let $\mathbf{u = \sqrt{n}}$. This implies $\mathbf{n = u^2}$. Substituting these into the inequality gives:

$$\mathbf{0.02u^2 - 1.1619u + 0.5 \geq 0}$$

' This gives two roots for $u$:

1. $u_1 = \frac{1.1619 - 1.1445}{0.04} \approx 0.435$
2. $u_2 = \frac{1.1619 + 1.1445}{0.04} \approx 57.66$

Since the parabola $0.02u^2 - ...$ opens upwards, the inequality is $\geq 0$ (true) *outside* its roots. Thus, the solution is $u \leq 0.435$ or $u \geq 57.66$.

Now, we substitute back $\mathbf{u = \sqrt{n}}$:

- $\sqrt{n} \leq 0.435 \implies n \leq 0.189$. (This is an extraneous root and not a valid sample size).
- $\sqrt{n} \geq 57.66 \implies n \geq (57.66)^2 \implies \mathbf{n \geq 3324.67}$

**Answer**

The inequality $n \geq 3324.67$ tells us the minimum $n$ required. Since $n$ must be a whole number of voters, we must round **up** to the next integer.

Minimum $\mathbf{n = 3325}$.

11. Let $X_1, \ldots, X_n$ be independent random variables with $X_i \sim \text{Uniform}(0, 1)$. Define

$$\widehat{I}_n = \frac{1}{n} \sum_{i=1}^{n} g(X_i), \qquad g(x) = e^x.$$

Let $I = \mathbb{E}[g(X)]$. Find a sample size $n$ such that

$$\mathbb{P}\big(|\widehat{I}_n - I| < 0.01\big) \geq 0.95.$$

(Hint: For a normal variable $Z \sim \mathcal{N}(0, \sigma^2)$, $\mathbb{P}(|Z| < 2\sigma) = 0.95$)

**Solution:**

**1. Mean of $g(X)$.**

$$I = \mathbb{E}[e^X] = \int_0^1 e^x \, dx = e - 1.$$

**2. Variance of $g(X)$.**

$$\mathbb{E}[e^{2X}] = \int_0^1 e^{2x} \, dx = \frac{e^2 - 1}{2},$$

so

$$\sigma^2 := \mathrm{Var}(g(X)) = \mathbb{E}[e^{2X}] - (\mathbb{E}[e^X])^2 = \frac{e^2 - 1}{2} - (e - 1)^2.$$

Numerically this is $\sigma^2 \approx 0.2420356$, hence $\sigma \approx 0.4919711$.

**3. Use the CLT approximation and the given normal fact.** By the central limit theorem, for large $n$ the distribution of $\widehat{I}_n - I$ is approximately normal with variance $\sigma^2/n$. Using the provided fact for a normal variable,

$$\mathbb{P}\big(|\widehat{I}_n - I| < 0.01\big) \approx \mathbb{P}\Big(\big|Z\big| < \frac{0.01}{\sigma/\sqrt{n}}\Big) \overset{!}{\geq} 0.95,$$

and the fact $\mathbb{P}(|Z| < 2\sigma) = 0.95$ implies we require

$$\frac{0.01}{\sigma/\sqrt{n}} \geq 2,$$

i.e.

$$0.01 \geq 2\frac{\sigma}{\sqrt{n}} \quad \Longrightarrow \quad \sqrt{n} \geq \frac{2\sigma}{0.01} \quad \Longrightarrow \quad n \geq \left(\frac{2\sigma}{0.01}\right)^2.$$

**4. Evaluate numerically.** Using $\sigma \approx 0.4919711$,

$$n \geq \left(\frac{2 \times 0.4919711}{0.01}\right)^2 \approx 9681.42.$$

Rounding up to an integer gives

$$\boxed{n = 9682}$$

samples.

12. Suppose $X_1, \ldots, X_n \sim$ iid (continuous) Uniform$[0, 1]$. Let

$$Y_n = e^{X_1} + \cdots + e^{X_n}.$$

1. What is the range of values taken by $Y_n$?
2. Find $E[Y_n]$ and $\mathrm{Var}(Y_n)$.
3. Approximate $P(|Y_n - E[Y_n]| > \dfrac{n}{2})$ using Chebyshev's inequality.
4. Approximate $P(Y_n > 2n)$ using the CLT.

**Solution:** Let $X_1, \ldots, X_n \sim$ iid Uniform$[0,1]$. Define

$$Y_n = e^{X_1} + e^{X_2} + \cdots + e^{X_n}.$$

(a) **Range of Values Taken by $Y_n$**

Since $X_i \in [0,1]$, we have:

$$n \leq Y_n \leq ne.$$

Thus, the range of $Y_n$ is $[n, ne]$.

(b) $E[Y_n]$ **and Var**$(Y_n)$

1. $E[Y_n] = n(e-1)$.
2. $\text{Var}(Y_n) = n\left(\frac{e^2-1}{2} - (e-1)^2\right)$.

(c) **Approximate** $P(|Y_n - E[Y_n]| > n^2)$ **using Chebyshev's Inequality**

By Chebyshev's inequality,

$$P(|Y_n - n(e-1)| > n^2) \leq \frac{\frac{e^2-1}{2} - (e-1)^2}{n^3}.$$

(d) **Approximate** $P(Y_n > 2n)$ **using the CLT**

Using the CLT,

$$P(Y_n > 2n) \approx P\left(Z > \sqrt{n} \cdot \frac{2-(e-1)}{\sqrt{\frac{e^2-1}{2} - (e-1)^2}}\right).$$